

Aryan Ganju

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PROFESSIONAL SUMMARY

Computer Science undergraduate at the National University of Singapore with experience building AI-enabled backend systems and production software in enterprise trading and large-scale application environments. Skilled in designing scalable services, data processing pipelines, and reliable automation to improve real-world operational workflows.

EDUCATION

National University of Singapore Aug 2022 – May 2026
Bachelor of Computer Science — Specializations: Artificial Intelligence & Database Systems
• Relevant Coursework: Algorithms, Artificial Intelligence/Machine Learning, Database Systems

EXPERIENCE

Mercuria Asia Resources Pte Ltd May 2025 – Oct 2025
Software Engineering Intern
• Engineered an **AI/NLP trade validation pipeline** reconciling unstructured trade inputs with structured datasets, achieving **97% accuracy** and reducing manual workload by **80%** across Middle Office workflows.
• Deployed the system as a **scalable FastAPI microservice with asynchronous batch processing**, cutting validation turnaround time by **70%** and enabling seamless integration into enterprise trading infrastructure.
• Built a **dynamic clause library with semantic matching, normalization, and RAG-assisted validation**, reducing recurring false mismatches by **35%** while improving legal consistency across commodity contracts.
• Developed internal tooling to transform **raw fixed-derivative trade inputs into structured, analyst-friendly formats**, reducing interpretation errors by **40%** and accelerating downstream reconciliation processes.

TVS Digital Pte Ltd Jun 2024 – Dec 2024
Software Engineer Intern
• Automated QA data generation using **Python and MySQL** and introduced **test-driven development with 300+ unit tests**, increasing coverage to **95%** and preventing **five production defects** prior to release.
• Led end-to-end **Viber API integration** across **seven international partners**, delivering a reliable cross-platform customer messaging capability within the production ecosystem.
• Improved **AWS-based deployment reliability and system efficiency** through monitoring and optimization scripts, reducing deployment time by **20%** and lowering compute costs by **15%**.

PROJECTS

GUI Murphy (TikTok TechJam) Sep 2025
• Built an **AI-driven GUI testing assistant** capable of detecting subtle UI state changes such as color shifts and icon toggles that are typically missed by traditional automated testing tools.
• Designed a **multimodal computer-vision and reasoning pipeline** combining YOLO, CLIP, and GPT-4 with **FastAPI and WebSockets** to enable real-time visual comparison and developer feedback.
• Ranked **Top 12 of 200+ teams** at **TikTok TechJam 2025**.

SKILLS

• **Languages:** Python, Java, TypeScript, JavaScript, C, HTML, CSS, SQL
• **Frameworks & Libraries:** React, Angular, Spring Boot, Node.js, Flutter, SQLAlchemy, PyTorch, Jest, React Testing Library
• **Databases:** PostgreSQL, MySQL, MongoDB, Supabase
• **Cloud & Tools:** AWS, GCP, Azure, Docker, Git, Hadoop, Spark, Vercel, Inngest, Arcjet, Redis, NoSQL, Kafka
• **Methodologies:** Agile, Test-Driven Development (TDD), Event-Driven Architecture

LEADERSHIP & ACTIVITIES

Head of Operations, Google Developer Student Club (NUS) Aug 2024 – Aug 2025
• Scaled engagement across a **200+ member developer community**, increasing participation by **30%** through redesigned event formats and organizing the **Hack for Good** hackathon with **600+ participants**.

Volunteer, Coastal Cleanup (Singapore) Aug 2022 – Aug 2024
• Contributed to environmental sustainability through recurring coastal clean-up initiatives and community participation.